linearization and stability

Jacobian matrix

the Jacobian matrix of a vector field at any (x, y) is:

$$A = \begin{bmatrix} \frac{\partial x'}{\partial x} & \frac{\partial x'}{\partial y} \\ \frac{\partial y'}{\partial x} & \frac{\partial y'}{\partial y} \end{bmatrix}$$

here you need to find the critical points they aren't $(0,0)$ anymore

$x' = -x + x^3$ Find all fixed point and classify them

$y' = -2y$

let $x' = 0 \Rightarrow x(-1 + x^2) = 0 \longrightarrow x = 0, \quad -1 + x^2 = 0$

$x^2 = 1$

$x = \pm 1$

let $y' = 0 \Rightarrow -2y = 0$

$y = 0$

Critical points $= (0,0), (1,0), (-1,0)$

you need to apply the Jacobian matrix

$$A = \begin{bmatrix} -1 + 3x^2 & 0 \\ 0 & -2 \end{bmatrix}$$

now we Plug in all the critical Points into the Jacobian matrix

@ $(0, 0)$ $\Longrightarrow$ $A = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$

$\tau = -1 + 2 = 1$

$\Delta = (-1)(2) - (0)(0) = -2$

$$\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2} = \frac{(1) \pm \sqrt{(1)^2 - 4(-2)}}{2} = \frac{1 \pm 3}{2} \qquad \begin{matrix} \lambda = 2 \\ \lambda = -1 \end{matrix}$$

stable node

@ $(1, 0)$ $\Longrightarrow$ $A = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$

$\tau = 2 - 2 = 0$

$\Delta = (2)(-2) - (0)(0) = -4$

$$\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2} = \frac{(0) \pm \sqrt{(0)^2 - 4(-4)}}{2} = \frac{4}{2} \qquad \begin{matrix} \lambda_1 = 2 \\ \lambda_2 = 2 \end{matrix}$$

degenerate stable Node

@ $(-1,0)$ $\Rightarrow$ $A = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$

$\tau = 2 - 2 = 0$

$\Delta = (2)(-2) - (0)(0) = -4$

$$\lambda = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2} = \frac{(0) \pm \sqrt{(0)^2 - 4(-4)}}{2} = \frac{4}{2} \qquad \begin{matrix} \lambda_1 = 2 \\ \lambda_2 = 2 \end{matrix}$$

degenerate stable Node